

- 6 The best time to obtain a clear image of chromosomes during a mitotic cell cycle is during the metaphase stage.

Fig. 6.1 is a scanning electron micrograph of a group of human chromosomes at metaphase.

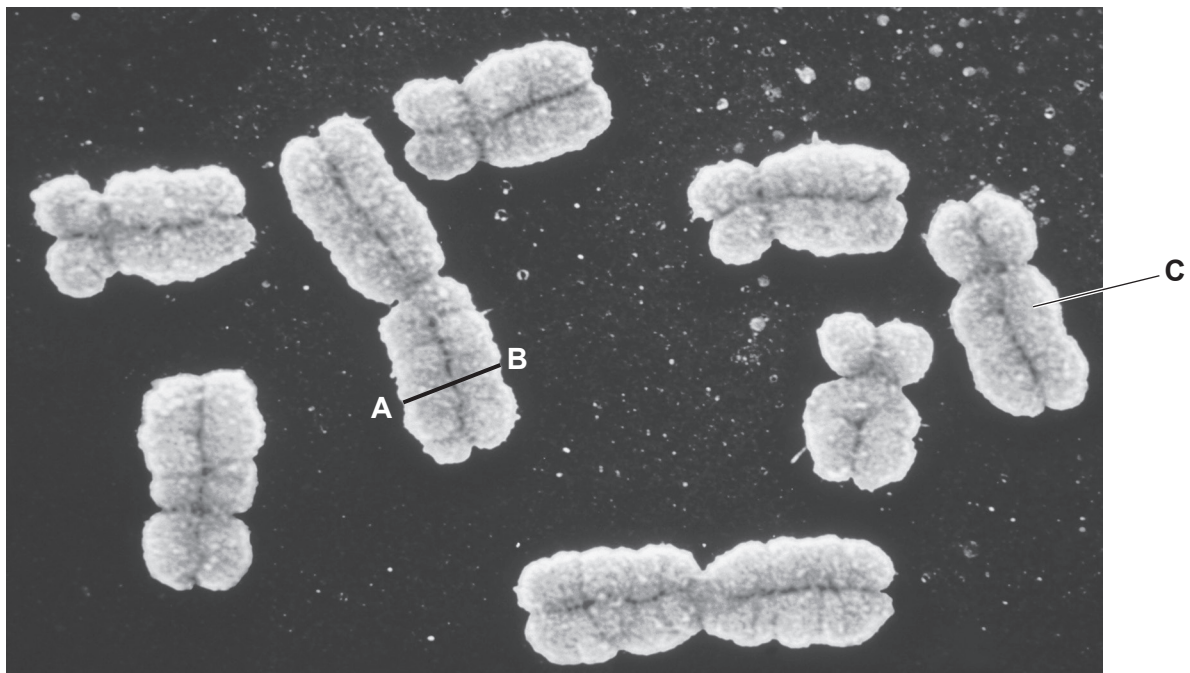


Fig. 6.1

- (a) The group of chromosomes shown in Fig. 6.1 is magnified many times.

The actual width of the human chromosome between **A—B** is 1400 nm.

Calculate the magnification of the scanning electron micrograph shown in Fig. 6.1.

magnification × [2]

(b) Draw chromosome **C** in Fig. 6.1 in the space provided.

Label your drawing to show the structure of the chromosome.

[4]

(c) Suggest why the metaphase stage is the best time during a mitotic cell cycle to obtain a clear image of chromosomes.

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..... [1]

(d) Name the stage of mitosis that immediately follows metaphase.

..... [1]

[Total: 8]